

GymOS ERP v2 Documentation Summary (Part 1 → Part 10)

These 10 parts are **not the application itself**. They are the **Product Requirement Document (PRD) + Software Architecture + UI/UX Specification + Engineering Blueprint** that your development team (or AI coding agent) will use to build the application.

Think of it like this:



Without documentation, developers constantly ask questions and make inconsistent decisions. With a detailed specification, everyone builds toward the same vision.

Part 1 — Product Foundation

Purpose

This defines **what GymOS is**.

Contains

- Product Vision
- Executive Summary
- Business Goals
- User Roles
- Core Modules
- Functional Requirements (FR-001-100)
- User Stories (US-001-070)
- Architecture Principles

Why it's useful

This answers:

Why are we building this product?

Without this, every developer imagines a different product.

Part 2 — Product Structure

Purpose

Defines **how the product is organized**.

Contains

- Information Architecture
- Navigation
- Business Rules
- FR-101-200
- US-071-140
- Screen Inventory

Why it's useful

Instead of random pages, it creates a logical application structure.

For example:

Dashboard

↓

CRM

↓

Members

↓

Billing

↓

Inventory

↓

Everything is connected.

Part 3 — User Experience

Purpose

Explains

| How users move through the system.

Contains

- Member Lifecycle
- AI Workflow
- User Flows
- Reception Flow
- Owner Dashboard
- Mobile Structure

Why it's useful

Instead of thinking

| What screen comes next?

the developer already knows.

Example

Lead

↓

Trial

↓

Membership

↓

Attendance

↓

Part 4 — Database Architecture

Purpose

Defines the application's data model.

Contains

- PostgreSQL structure
- Multi-tenancy
- Core Tables
- Relationships
- RBAC
- Audit Logs
- Index Strategy

Why it's useful

Everything depends on the database.

If the database is wrong,

the whole application becomes difficult to maintain.

Part 5 — Backend APIs

Purpose

Defines communication between frontend and backend.

Contains

- REST APIs
- Authentication
- Payment Flow
- WhatsApp Flow
- Events
- Error Codes

Why it's useful

Frontend developers know

GET /members

POST /attendance

POST /payments

Backend developers know exactly what to build.

Part 6 — Design System

Purpose

Makes the UI consistent.

Contains

- Colors
- Typography
- Spacing
- Buttons
- Cards
- Inputs
- Component Library
- Accessibility

Why it's useful

Instead of

100 different button styles

you get

1 standard.

That makes the application look professional.

Part 7 — Desktop UX

Purpose

Designs the desktop application.

Contains

- Dashboard Layout
- CRM Screen
- Member Screen
- Billing
- Reports
- AI Panel
- Interaction Rules

Why it's useful

The frontend team knows

where everything goes.

No guessing.

Part 8 — Mobile UX

Purpose

Designs mobile apps.

Contains

Member App

Trainer App

Reception App

Owner App

Notifications

Offline Mode

Responsive Rules

Why it's useful

Instead of shrinking the desktop UI,

the mobile app is designed separately.

This is a huge improvement.

Part 9 — AI Platform

Purpose

Makes the software intelligent.

Contains

AI

Automation

CRM Intelligence

Predictions

Revenue Forecast

Workout AI

Diet AI

Business Intelligence

Why it's useful

Instead of software showing

Revenue

AI says

Revenue will drop next month because renewals are decreasing.

This is the future.

Part 10 — Engineering

Purpose

Shows developers

how to build the application.

Contains

Technology Stack

Folder Structure

Microservices

Docker

CI/CD

Testing

Security

Roadmap

Developer Handoff

Why it's useful

Developers can start coding immediately.

Overall Architecture

Together these 10 parts define the entire system:

```
graph TD; A[Product Vision] --> B[Business Rules]; B --> C[UX]; C --> D[Wireframes]; D --> E[Database]; E --> F[Backend APIs]; F --> G[Frontend]; G --> H[Mobile]; H --> I[AI]; I --> J[Deployment];
```

Product Vision
|
Business Rules
|
UX
|
Wireframes
|
Database
|
Backend APIs
|
Frontend
|
Mobile
|
AI
|
Deployment

Who Uses Each Part?